Prepare Abiotic Data for Model Fitting

Description

Reshapes long-format abiotic data into a wide data frame suitable for downstream scaling, filtering to the canonical sample index supplied by 'align_sample_ids()'.

Usage

```
prepare_abiotic_for_fit(data_abiotic_long = NULL, data_sample_ids = NULL)
```

Arguments

`data_abiotic_long`	A data frame in long format with columns 'dataset_name', 'age', 'abiotic_variable_name', and 'abiotic_value'.
`data_sample_ids`	A data frame of valid '(dataset_name, age)' pairs as returned by 'align_sample_ids()'.

Details

Unlike the former 'prepare_data_for_fit(type = "abiotic")', this function retains 'dataset_name' and 'age' as real columns so that 'scale_abiotic_for_fit()' can operate on them without parsing row names. Row names are added by 'scale_abiotic_for_fit()' as part of the scaling step.

Value

A data frame in wide format with real columns 'dataset_name', 'age', and one column per abiotic variable. Rows are ordered by 'dataset_name' then 'age', matching the order in 'data_sample_ids'. Missing variable–sample combinations are left as 'NA'. The 'age' column is preserved as a plain numeric column (not encoded in row names) to simplify downstream scaling.

See Also

[align_sample_ids()], [scale_abiotic_for_fit()], [build_jsdm_fit_input()]